

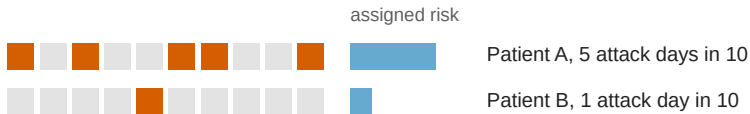
(a) AUROC is a statement about ordered pairs

The probability that a randomly drawn attack day is ordered above a randomly drawn quiet day, with tied pairs contributing one half:

$$\text{AUROC} = \frac{n_{\text{concordant}} + \frac{1}{2} n_{\text{tied}}}{n_{\text{pairs}}}$$

Which pairs are admitted to that average is a choice, and it is the choice separating the two columns of this paper's results.

(b) Worked cohort: a classifier using patient identity alone



One risk per patient, held constant across that patient's days: no day-level information is present.

(c) Pooled: pairs drawn from the cohort as a whole



$n_{\text{pairs}} = 84$: 45 concordant and 5 discordant, each pairing days from different patients; 34 tied, each pairing days from one patient

$$\text{AUROC} = \frac{45 + \frac{1}{2}(34)}{84} = 0.74$$

(d) Within person: pairs drawn inside each patient



$$C = \frac{0 + \frac{1}{2}(34)}{34} = 0.50$$